Kyle Owen Rex

***Mechatronics (Electrical, Mechanical, Software) Engineer | Active Secret Level US Security Clearance* | 2+ Years Relevant Experience**
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Educational and Organizational Experience

Texas A&M University College of Engineering - (TAMU) - College Station, TX - (August 2022 - Present) - (Graduate May 2026)

- GPA: 3.4 (125 credit hours completed) (Unofficial Transcript on LinkedIn).
- B.S. - Multidisciplinary Engineering Technology (ABET Accredited) (Electrical, Mechanical, Electronics, and Software Engineering).
- Major in Mechatronics (MXET) and Minor in Embedded Systems Integration (EMSI).
- Mechatronics program emphasizes the integration of mechanical, electrical, electronic, and software engineering, combining elements of robotics, automation, control systems, computer science, embedded systems, computer aided design, telecommunications, and AI/ML.
- (1) College of Engineering Deans Honor Roll Award for Outstanding Academic Achievement for a Semester GPA >3.75.
- (3) College of Engineering Distinguished Student Awards for Outstanding Academic Achievement for a Semester GPA >3.50.
- Former Peer Notetaker for Texas A&M Disability Services - took notes for students with disabilities.
- Sandia Sponsored Texas A&M Aggies Invent 48-hour Design Competition. Awarded 1st place prize, most feasible solution, best presentation.
  - Developed and presented a proof of concept and prototype for the integration of conductive nanowires with RFID technology into the packaging of assets being transported to provide real time tamper monitoring. Texas A&M Engineering Article featured in (on LinkedIn).
- Professional Institute of Electrical & Electronics Engineers (IEEE) Technology Member at Texas A&M.
- Texas A&M Society of Mechatronics Engineering Technology (SOMTECH) Organization Mechanical Engineering Team Member.
  - Perform extensive R&D to generate plans for the physical development of a team Rover to participate in the University Rover Challenge.
- Selected by Texas A&M Cybersecurity Center to participate in the Department of Defense Cyber Leadership Development Program (CLDP).
  - 2025 Joint Service Academy Cybersecurity Symposium (JSACS) Attendee at the USMA Westpoint.
- Texas A&M Corps of Cadets Cyber Operations Special Unit Member. 2025 NSA Cyber Exercise (NCX) Forensics Module Team Leader.

Lassiter High School - (LHS) - STEM Engineering Academy PLTW 4 Year Program - Marietta, GA - (August 2018 - May 2022)

- Skills learned included Autodesk Inventor Autodesk AutoCAD/Drafting, 3-D Modeling/Printing, Sketchup, VEX Robotics/coding, Isometric Sketching, Aery, X-Plane/Microsoft Flight Simulation, Spreadsheet Data Analysis, Robot Design/Programming Code, and Presentation Skills.
- Engineering capstone project which was a heat-resistant grill camera capable of remote monitoring food being grilled using Bluetooth.

Work Experience

Sandia National Labs - (SNL) - (Subsidiary of Honeywell International) - (May 2024 - Present)

- Onsite Summer (Full-time 40-hour work weeks) and Year-Round (Part-time) Weapon Subsystems R&D Internship with the Livermore Lab (CA) at the George H.W. Bush Combat Development Complex Research Integration Center (RIC) (TX).
  - Developed testing procedures and enhanced understanding of Buck Converters with Type III Topology Compensators for Switch Mode Power Converters and DC-to-DC feedback-controlled conversion. This initiative aimed to transition conventional multi-tap battery systems to a modular power bus-based system improving applications in space missions, nuclear disaster response & radiation-hardened systems.
  - Created a method for obtaining Bode plots of live DC-to-DC feedback-controlled Switch Mode Power Converters using a Tektronix Mixed Signal Oscilloscope. This involved analyzing Bode plots from ideal MATLAB transfer functions, Simulink simulations, and physical circuit tests. The successful testing procedure and hardware toolset enabled the team to evaluate both in-development and existing converters.
  - Created LTspice simulations of the complete closed-loop buck converter system, which I replicated in a circuit breadboard design & tested.
  - Achieved proficiency in Mentor Graphics Xpedition for PCB design by completing two designs from concept to production, conducting design reviews to incorporate team feedback, performing design rule checks, & generating Gerber manufacturing files. Gained expertise in lab testing, solder rework, and general lab operations while successfully delivering high-level test objectives, plans, reports, and results.
  - Presented summer 2024 work in the form of a poster, report, and presentation at the CA Summer Intern Symposium (All on LinkedIn).
  - Sandia Lab News Article I was featured in regarding my active participation in career development events (on LinkedIn).
  - Onsite Mentor Reference: Dr. Matthew McDonough. Work Phone: 505-228-4550. Email: mmcdono@sandia.gov (preferred).
- Year-Round (Part-time) Fall 2025 - Spring 2026 Student Ambassador
  - Represented Sandia National Laboratories on campus by supporting recruiting events, career fairs, and information sessions while sharing personal internship experiences with students. Collaborated with recruiters to coordinate events, schedule career discussions, and promote opportunities to attract top talent.

National Aeronautics and Space Administration - (NASA) - (Independent Agency of U.S. Federal Government) - (January - May 2022)

- Remote Spring R&D Internship with the Ames Research Center (CA) at Lassiter High School (Part-time 10-hour work weeks).
  - Collaborated on a research project analyzing aeronautical contributions to wildfire management, focusing on the application of manned and unmanned vehicles for detection and suppression technologies. Developed a proof of concept for integrating programmed AI prediction software into satellites to enhance wildfire prediction, mitigation, & management by utilizing historical climate & weather data.

Textron Aviation - (TXTAV) - (Beechcraft and Cessna Conglomerate) - (July 2022)

- Onsite Civil Air Patrol Aircraft Manufacturing Academy Apprenticeship at Independence Kansas Piston Engine Aircraft Manufacturing Factory
  - Gained hands-on experience in manufacturing and quality assurance within aircraft production facilities for the Cessna 208, M-2, 172, 206, & 182. Responsibilities included installing AC units, starters, belt drives, gas lines, & oil lines on a 208 Turboprop engine; performing damage assessments and rivet installations on a 182's wings; executing detail work on a 206, including sanding, buffing, & paint discrepancy identification; inspecting electrical systems & insulation on an M-2; checking wheel torque on a 206; conducting quality checks on tools; performing upholstery tasks such as carpeting, grommeting, & sewing; along with welding, cutting, & drilling steel beams.

Part Time Online Sales Entrepreneur & Independent Financial Strategist / Advisor - (March 2020 - Present)

- Small business student owner, manager, & entrepreneur. Successfully operate a passive part time online business that at its peak had 1 part-time family employee. Currently over 2100 items sold with over $51,000 in gross sales since launch. This allowed me to educate myself in managing personal finances to the extent that I was awarded 1st Place on an essay contest for "how personal financial readiness will impact future military operational readiness". I have held multiple classes and discussions educating peers on proper financial management.

Engineering / Cyber Projects *(T) = Team | (A) = Alone | (L) = Report on LinkedIn | (G) = Code on GitHub page*

Los Alamos National Labs Vacuum/ Flexible Auger Capstone Project (MMET 429 & MMET 422, Manufacturing Technology Projects) (T)

- Positions include project management lead, electrical engineering & manufacturing lead, and requirements & verifications lead.
- Will design a vacuum/flexible auger material transport hose attachment for the Brokk 70 robot that will need to transport dry powder and material with the consistency of peanut butter while meeting any additional requirements from LANL.
- Project motivated by DOE need to remove 12 million gallons of "sludge" waste stored in the Hanford tanks in Washington State. Sludge is made up of water-insoluble metal oxides, with a significant amount of interstitial liquid, forming what the DOE office of River Protection describes as a peanut butter like texture. The mechanical properties of the waste complicate the use of mobile robots in this environment for the purpose of excavating and removing the waste using dry retrieval methods.

C/C++ Arduino Uno R3 IDE Ball and Beam PID Feedback Control System (MXET 375, Applied Dynamic Systems) (T) (L)

- Built a physical prototype and Simulink simulation based on its derived equation of motion with the goal of matching the results. The beam would move based on the ball's position until the ball reached an equilibrium state. I developed the entire prototype including the Arduino code and the MATLAB code to measure the results of the prototype.

Python Visual Studio Code SCUTTLE "Following Serving Robot" C/C++ Arduino Uno R3 IDE (MXET 300, Mechatronics I) (T) (L)

- Built to follow a person with obstacle avoidance using camera color tracking/lidar sensing. Also, refills their cup using a motion sensor, servomotor, and 3D printed cup holder/arm controlled using the Arduino. I developed the entirety of the pouring system including 3D printing the cup holder with my personal 3D printer then coding the Arduino to read a motion sensor to turn the arm with a servo motor.

Arm Assembly Keil uVision5 "Guess the Number" Game Nucleo-64 STM32 Controller (ESET 349, Microcontroller Architecture) (A) (L)

- Developed an interactive game using LEDs, motion sensors, and a 16x2 LCD screen. The game displays a prompt on the screen to "Guess the number" and the user toggles the sensors guessing which number sensor is right (green LED) or wrong (red LED).

CAD Autodesk Fusion 3D Printed Automated Pet Food Dispenser (MMET 361, Product Design and Solid Modeling) (T)

- Built using only 3D printed parts from CAD Autodesk Fusion and a servo motor focusing on mechanical tolerancing and dimensioning. I worked specifically on designing and printing the swinging motor arm that would turn to uncover a hole allowing the food to fall onto a ramp.

C/C++ Keil uVision5 MSP432 Microcontroller Programming Project (ESET 269, Embedded Systems Development) (A) (G)

- Developed a code sequence to measure and configure the inputs and outputs including the LEDs, Buttons, and Internal Temp.

Analog "Clapper Circuit" (ESET 350, Analog Electronics) (A) (L)

- Built using basic circuity components and a microphone that I soldered together. An LED turned on and off based on sound inputs.

C/C++ Code Composer Studio "Plant Monitoring System" MSP430FR5994 Controller (ESET 369, Embedded Systems Software) (A) (L)

- Created a proof of concept that utilized a LCD Screen, HC-05 Bluetooth device, capacitive soil moisture sensor, and LM35 temp sensor.

Python Pycharm Ticktacktoe Game with Machine Opponent Option (ENGR 102, Engineering Development in Python) (T) (G)

- Developed the code sequence and functions responsible for being the machine opponent. Can play against another player or the machine.

Engineering / Cyber Technical Skills

Electrical, Software, and Mechanical Engineering Software Skills

- Proficient in a wide range of software and tools, including Siemens Mentor Graphics Xpedition, MATLAB, Simulink, LTspice, PyCharm, Code Composer Studio, Thonny IDE, Spyder, Intel Quartus, Google Collaboratory, MobaXterm, Visual Studio Code, Keil uVision5, Oracle VM VirtualBox, Arduino IDE, Tera Term, AutoCAD, Autodesk Inventor, Autodesk Fusion, and Automation Studio. Skilled in coding languages such as Arm Assembly, Java, Python, Micro Python, and C/C++.

Electrical and Software Engineering Hardware Skills

- Experienced in utilizing a variety of electrical equipment and components, including Arduino, ST, Tektronix, Picotest, Digikey, Texas Instruments, Terasic, & Keysight tools, as well as DC power supplies, digital multimeters, oscilloscopes, function/waveform generators, mixed signal oscilloscopes, & DC electronic loads. Proficient in working with FPGA DE-10 Lite, MSP432, MSP430FR5944, Nucleo-64 STM32, Arduino Uno R3, Raspberry Pi Pico, Raspberry Pi, & SenseHat, alongside DC-to-DC Buck Converters & various basic circuitry components.

Mechanical Engineering Hardware Skills

- Proficient in using chemical, fluid, and mechanical testing equipment, including basic chemistry lab tools, fluid mechanics apparatus, universal testing machines, torsion test rigs, and strain indicators, to conduct various tests such as hardness, tensile, impact, metallography, heat treatment, Jominy, and cold work on metallic materials like stainless steels. Additionally, own and operate a Creality Ender 3 v3 SE for 3D printing applications.

Military and Leadership Experience

US Army Reserves - 340TH Chemical (CBRN) Company - Houston, Texas - (April 2024 - Present)

- Currently serving as a 09R (MOS) SMP Reservist. Upon graduation I will commission into the Reserves as a Second Lieutenant where I will serve 1 weekend a month & 2 weeks during the year while furthering my education or pursuing my fulltime long-term engineering career.

Texas A&M Corps of Cadets - (Collegiate Paramilitary Fraternal Organization) - (August 2022 - Present)

- Member of the Senior Military College's Corps of Cadets - 1st Brigade 1st Battalion L-1 Lonestar Company.
- Regular class day begins at 0500 (5am) and ends at 2200 (10pm). Physical Training weekday mornings from 0530 to 0700.
- Work in a very high intensity, stressful environment that is meant to prepare you for success in your career.
- L-1 Career Readiness Officer, Finance Officer, Operations Officer, Logistics Officer, Recruitment Officer, and Platoon Leader.
- Former member of the L-1 Bonfire crew. Participated in the creation and execution of the annual student led Bonfire celebration.

Texas A&M Army Reserve Officer Training Corps - (AROTC) – (U.S. Military Officer Commissioning Program) - (August 2022 - Present)
- Graduated from 35-day Advanced Camp at Fort Knox which involved intensive training, developing leadership, tactical skills, and teamwork through rigorous training & field operations, while enhancing decision-making & problem-solving abilities in high-pressure environments.
- Experience creating and briefing large scale comprehensive mission operation plans to command staff that can last for multiple hours.
- Squad Leader in command of teaching cadets basic Army skills, leading cadets in combat training operations, & conducting physical training.
- Participation in (7) Field Training Exercises, ranging from 3 to 7 days long, meant to test ability to lead and perform platoon & squad level operations (Positions include PL, PSG, SL, Weasel, RTO, & Medic) (Performed tactical Blackhawk flight operations).
- Extensive first aid and emergency medical technician training to handle any medical emergency, combat or non-combat related.
- Former member of Rudders Rangers special Army unit - specializes in small unit infantry tactics and teamwork.

Boy Scouts of America - (BSA) - (Youth Leadership Development Program) - (August 2010 - July 2023)
- Boy Scout Eagle Rank. Awarded for fulfilling leadership positions, service hours, merit badges, and planning service projects.
- Attended NYLT (National Youth Leadership Training) weeklong leadership training program (Group/ Patrol Leader).
- Positions included Troop/ Patrol Leader, Troop New Scout Guide, and Assistant Scout Master.

US Air Force Auxiliary Civil Air Patrol - (CAP) - (Youth Military Leadership Development Program) - (August 2021 - May 2023)
- Cadet Technical Sergeant. This program helped develop my leadership ability and Aerospace/ Aeronautical Engineering knowledge.
- Georgia Wing Week-long Summer Encampment at Air National Guard base, Air Dominance Center. Awarded NCO of the Encampment.
- Path towards Private Pilot’s License. FAA (Federal Aviation Administration) Certified Student Pilots License. FAA 1st class medical certificate. Orientation flights in a single engine Cessna aircraft and glider. FAA Certified Recreational Drone Certification (TRUST).

Navy Junior Reserve Officer Training Corps - (NJROTC) - (Youth Military Leadership Development Program) - (May 2021 - May 2022)
- Class Officer and Commander. This program helped develop my leadership ability and general Military knowledge.
- Graduated from US Coast Guard Academy AIM 2021 summer program (Modeled Coast Guard basic training) onsite of USCGA.
- Member of the Sons of the American Legion and Georgia Boys State 2021 Graduate. Elected Commissioner of Labor.